

EP 402P: Engineering Physics Practicum

Course Code	EP 402P
Course Name	Engineering Physics Practicum
L-T-P-C	1-0-5-4
Prerequisites	PH301 or PH513 and PH501 or PH523
Intended for	B.Tech.
Distribution	Core
Approval	20th Senate (Apr 2019), in the B.Tech Engineering Physics curriculum; 37th BoA (Oct 2020); 30th Senate (Feb 2021), in the B.Tech Engineering Physics curriculum

Course Contents

Part-A

- **Hall Effect in Semiconductor**

Measure the resistivity and Hall voltage of a semiconductor sample as a function of temperature and magnetic field. The band gap, the specific conductivity, the type of charge carrier and the mobility of the charge carriers can be determined from the measurements. [Lectures]

- **Diffraction of ultrasonic waves**

Fraunhofer and Fresnel diffraction and determine the wavelength of the ultrasound wave. Students may also develop their own sound wave generator and receiver and perform this experiment. [Lectures]

- **Geiger-Muller-Counter**

To study random events. Determination of the half-life and radioactive equilibrium. Verification of the inverse-square law for beta and gamma radiation. [Lectures]

- **Dispersion and resolving power of a grating**

Determination of the grating constant of a Rowland grating based on the diffraction angle (up to the third order) of the high intensity spectral lines. Determination of the angular dispersion and resolving power of a grating. Students may also design). their own gratings and study that using this experimental technique. [Lectures]

Devices Fabrication Technologies

- **Basic clean room training and introduction to instruments**

Draw a comprehensive wafer clean process flow/cleaning of wafer and validate the hydrophobic and hydrophilic nature through contact angle measurements.

- **Metal-Semiconductor contact fabrications and characterizations**

Design and fabricate basic metal semiconductor junction and do the characterization. Basic characterization may also be extended to thin films grown via spin coater or any other technique.

Text Books

1. R. A. Dunlop, **Experimental Physics**, Oxford University Press (2012).
2. S. K. Ghandhi, **VLSI Fabrication Principles: Silicon and Gallium Arsenide**, 2nd Edition. (2014)

References

1. A. C. Melissinos, **Experiments in Modern Physics**, Academic Press (1996).
2. E. Hecht, **Optics**, 4 edition, Addison-Wesley. (2011)
3. J. Varma, **Nuclear Physics Experiments**, New Age Publishers (2010)
4. B. L. Worsnop and H. T. Flint, **Advanced Practical Physics for Students**, Methusen & Go. (1950).
5. E. V. Smith, **Manual for Experiments in Applied Physics**, Butterworths (1970).
6. D. Malacara (ed), **Methods of Experimental Physics**, Series of Volumes, Academic Press Inc. (1988).
7. D. K. Schroder, **Semiconductor Material and Device Characterization**, 3rd Edition.